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Linux Monitoring and Data Files Commands (Laboratory Exercise 3)

This laboratory/lesson is provided by our teacher.

Objectives:

- Utilize commands to view and manage running processes.
- Monitor disk space usage.
- Work with data files for viewing, sorting, searching, and compression.

Prerequisites:

Basic understanding of Linux commands
Access to a Linux system with administrative privileges.

Materials:

Terminal window
Text editor
“Music” directories created from the previous lab

Remember:

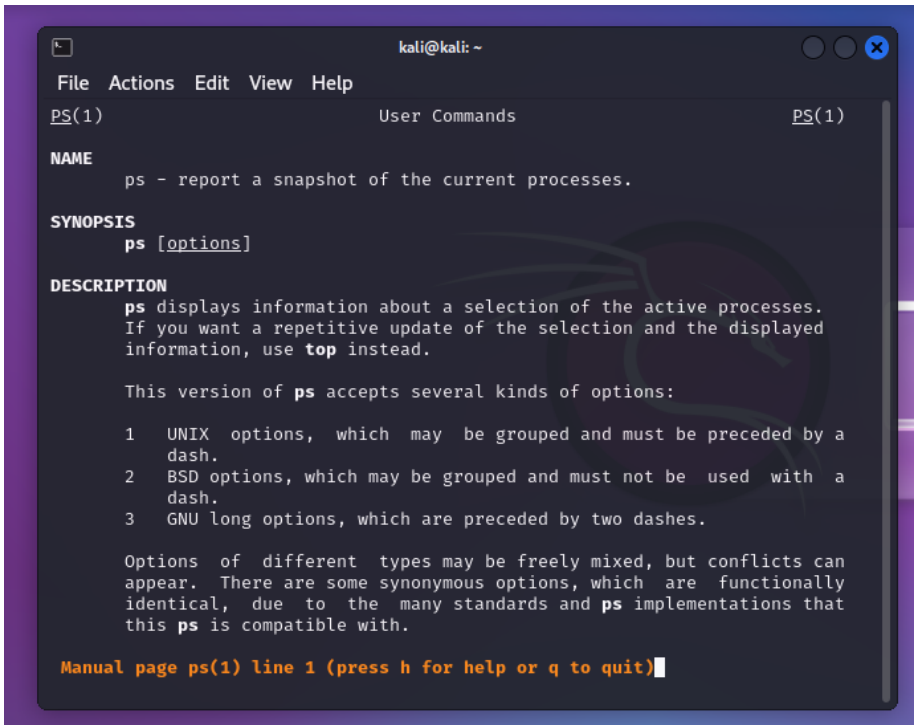
- Always be cautious when using commands with administrative privileges (sudo).
- Consult the manual pages (e.g., `man ps`) for detailed information and options for each command.

Tasks:

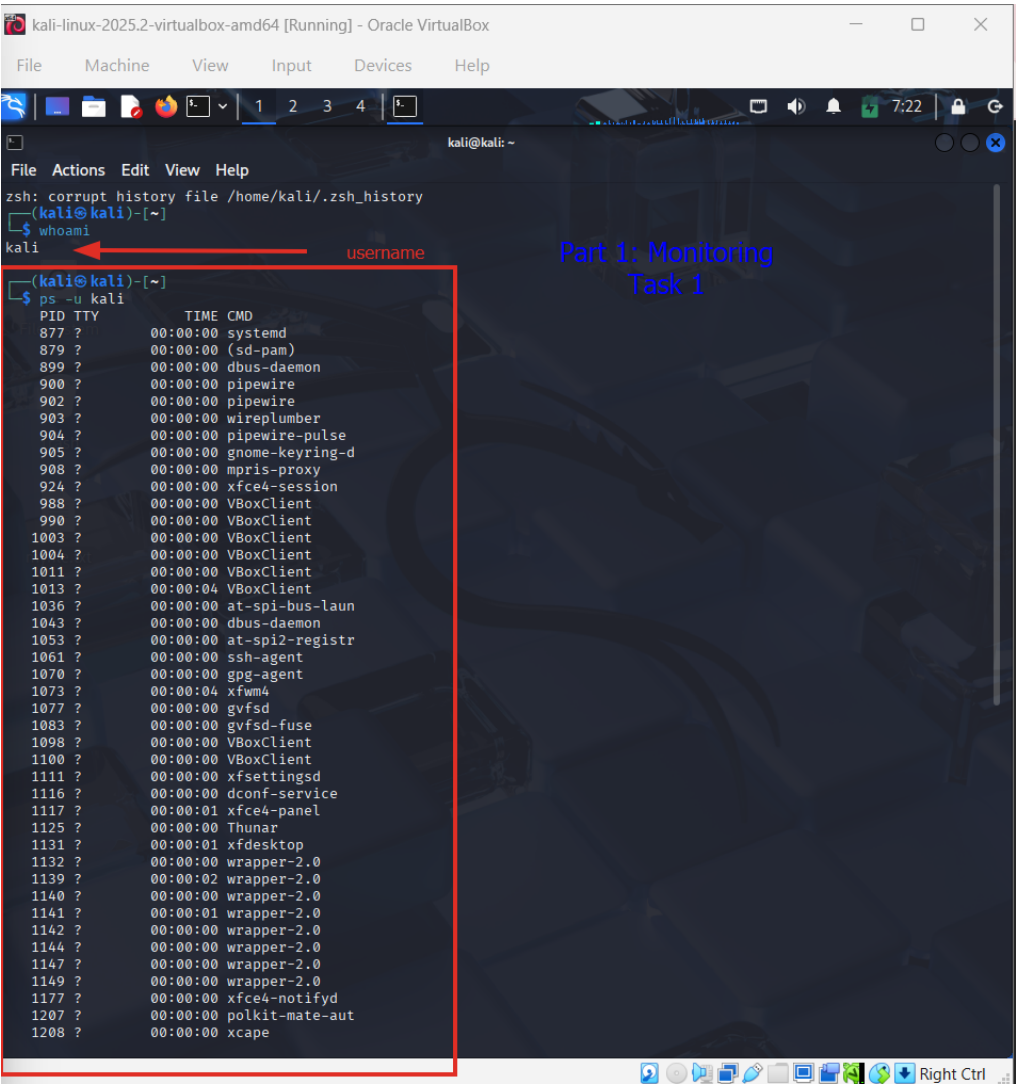
Part 1: Monitoring Programs

1. **Process Listing:** Use the `ps` command to list all running processes. Explore the various options for `ps`:

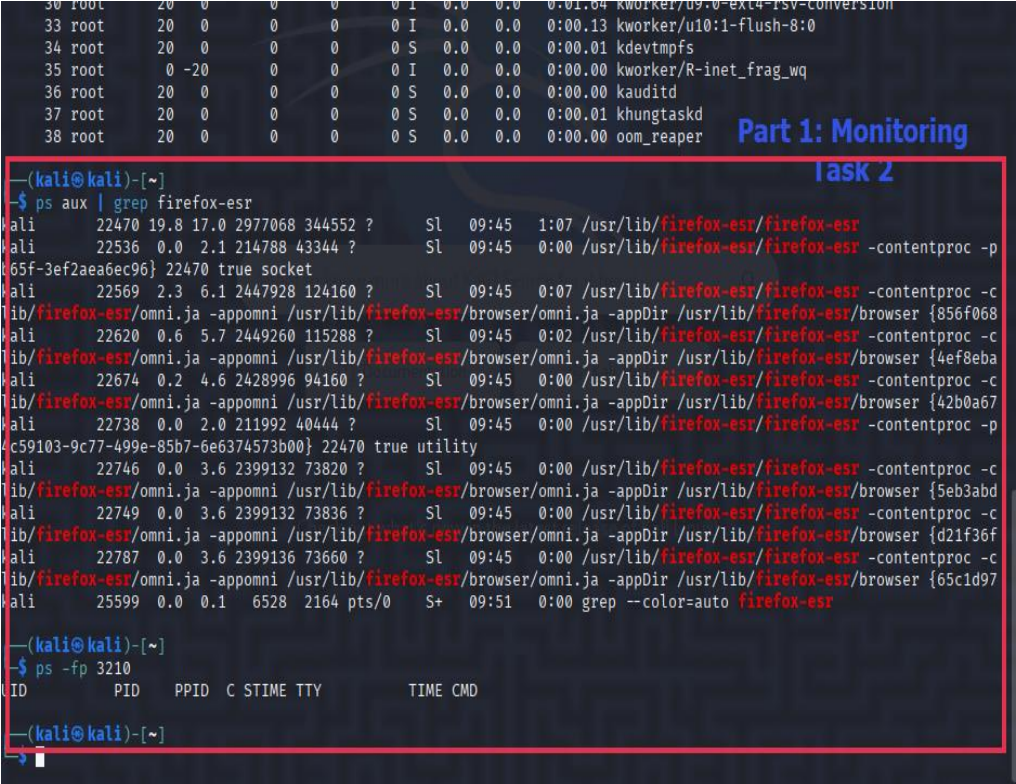
- `ps aux`: View detailed information about all processes.
- `ps -ef`: Display all processes, including those with short names.
- `ps | grep <program_name>`: Filter processes by name (e.g., `ps | grep firefox`).



Task 1: List all processes owned by a specific user (e.g., ps -u <username>).



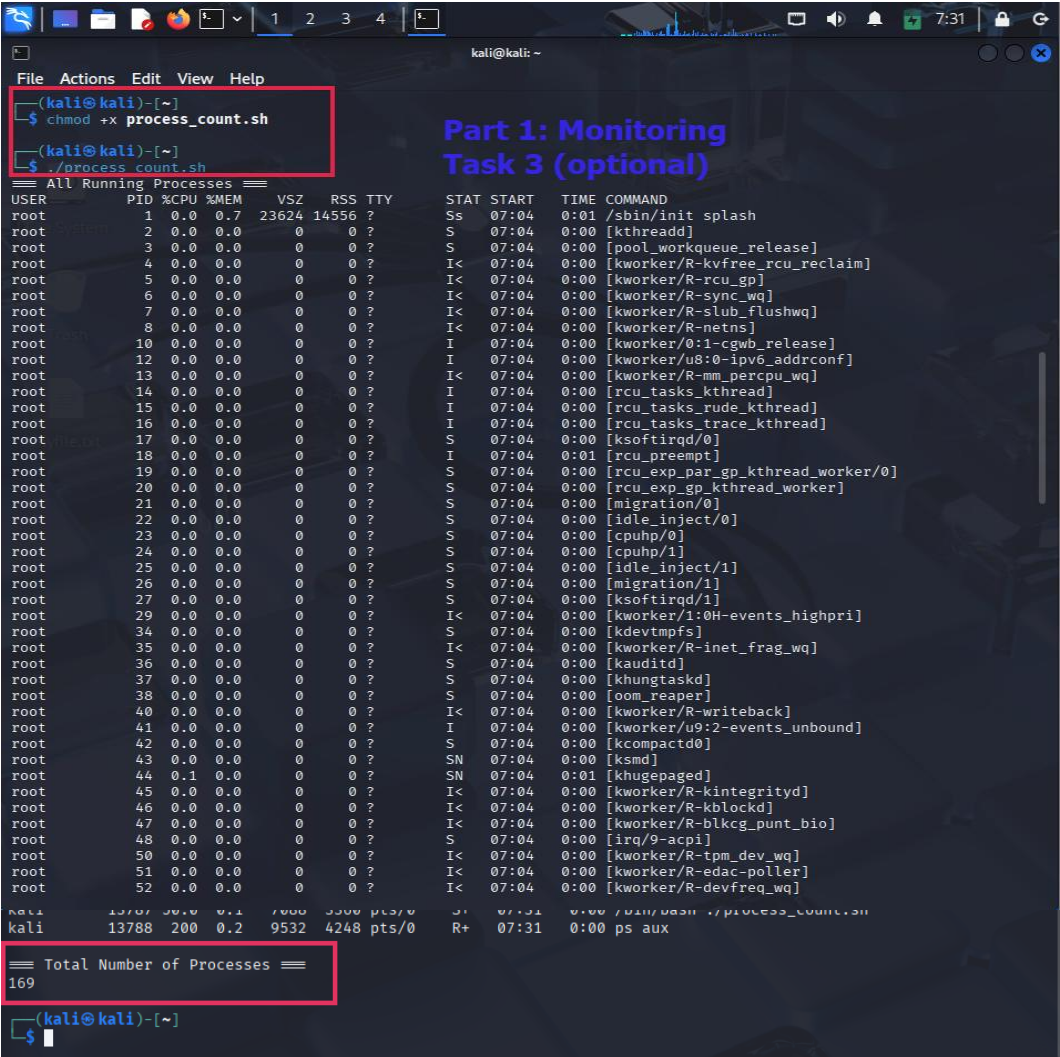
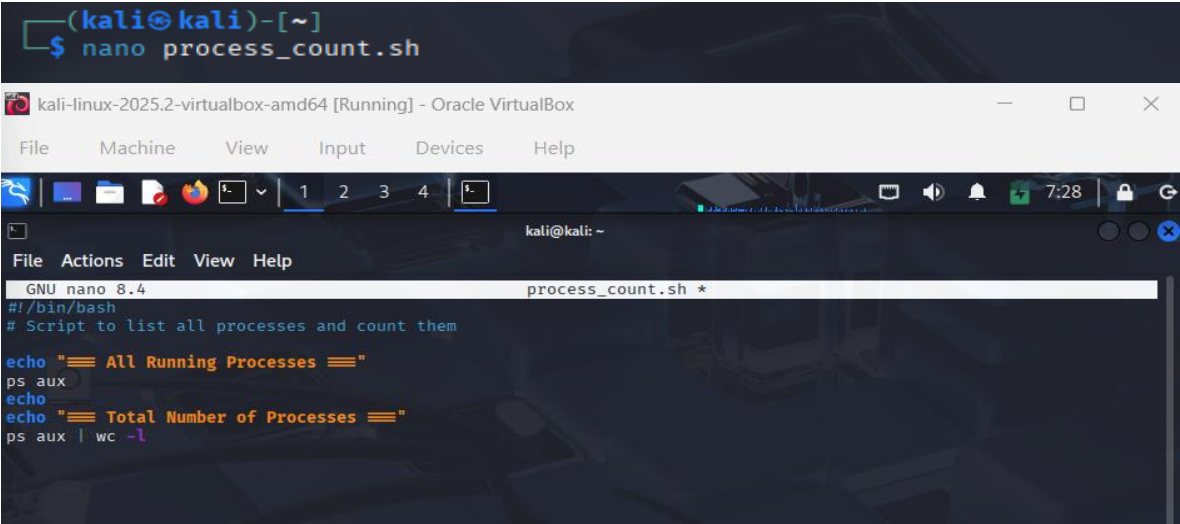
Task 2: Find the process ID (PID) of the program named "firefox" and display its detailed information.



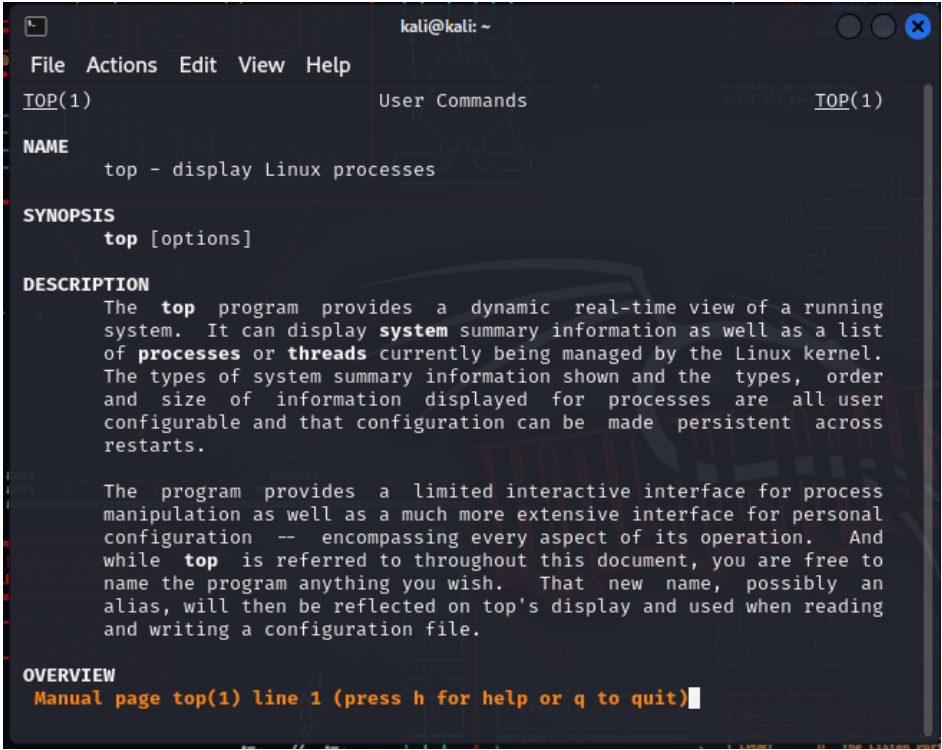
Here are the PIDs of the running firefox-esr processes:

22470
22536
22569
22620
22674
22738
22749
22787
22793

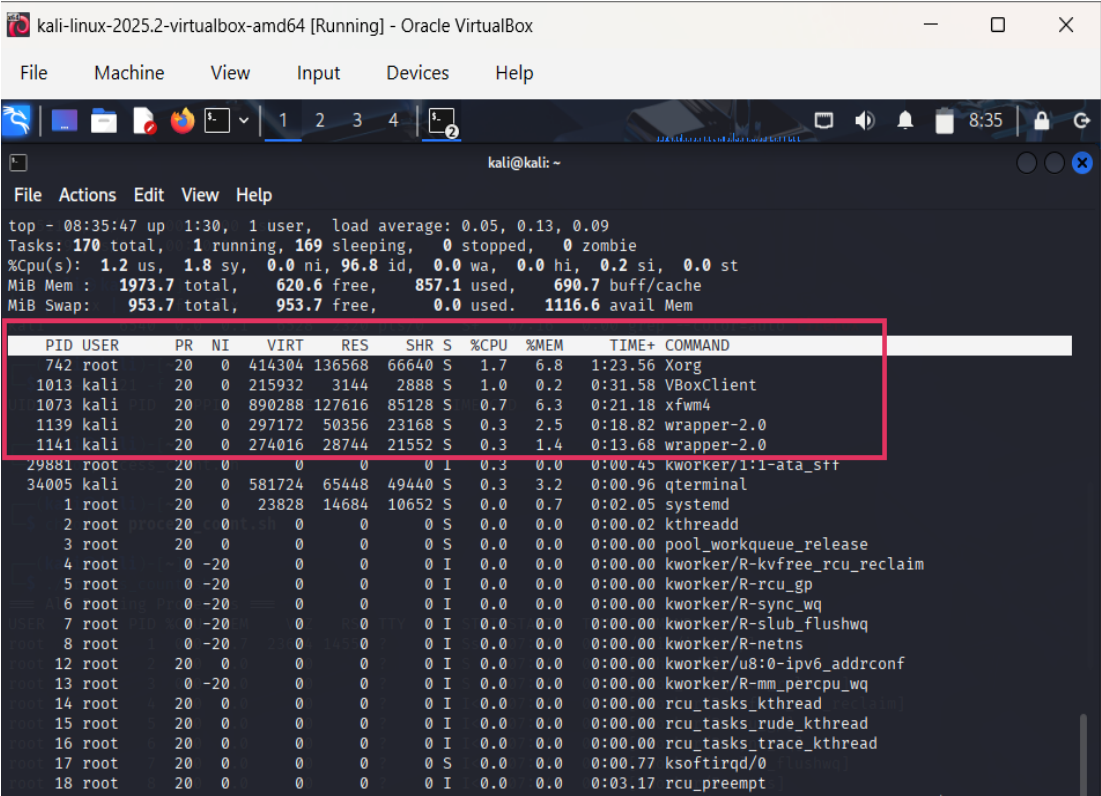
Task 3 (Optional): Explore writing a script that lists all processes and outputs the total number of processes running.



2. **Monitoring Resource Usage:** Use the `top` command to view real-time information about CPU, memory, and other system resources used by running processes. Analyze the information displayed and identify processes consuming high resources.



Task 1: Identify the top 5 processes consuming the most CPU resources.



Task 2: Monitor system activity for 5 minutes and observe how resource usage changes.

File

Actions

Edit

View

Help

top - 08:46:10 up 1:41, 1 user, load average: 0.57, 0.40, 0.22

Tasks: 171 total, 1 running, 170 sleeping, 0 stopped, 0 zombie

%Cpu(s): 1.6 us, 0.7 sy, 0.0 ni, 97.7 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 1973.7 total, 661.5 free, 815.1 used, 688.5 buff/cache

MiB Swap: 953.7 total, 953.7 free, 0.0 used. 1158.6 avail Mem

BEFORE

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1073	kali	20	0	890288	127616	85128	S	0.7	6.3	0:25.50	xfwm4
543	message+	20	0	10252	6488	4184	S	0.3	0.3	0:03.55	dbus-daemon
924	kali	20	0	347964	35108	18192	S	0.3	1.7	0:01.85	xfce4-session
1117	kali	20	0	509412	47468	33944	S	0.3	2.3	0:05.57	xfce4-panel
1139	kali	20	0	297172	50356	23168	S	0.3	2.5	0:22.11	wrapper-2.0
1141	kali	20	0	274016	28744	21552	S	0.3	1.4	0:16.38	wrapper-2.0
1147	kali	20	0	286492	38788	30852	S	0.3	1.9	0:04.85	wrapper-2.0
31849	root	20	0	0	0	0	I	0.3	0.0	0:01.12	kworker/1:0-events
34005	kali	20	0	581728	66876	49844	S	0.3	3.3	0:06.73	qterminal
34048	kali	20	0	10476	5792	3616	R	0.3	0.3	0:01.88	top
1	root	20	0	23828	14684	10652	S	0.0	0.7	0:02.14	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.02	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq

kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

top - 08:53:30 up 1:48, 1 user, load average: 0.08, 0.12, 0.15

Tasks: 169 total, 2 running, 167 sleeping, 0 stopped, 0 zombie

%Cpu(s): 1.4 us, 1.8 sy, 0.0 ni, 96.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st

MiB Mem : 1973.7 total, 626.0 free, 850.5 used, 688.6 buff/cache

MiB Swap: 953.7 total, 953.7 free, 0.0 used. 1123.2 avail Mem

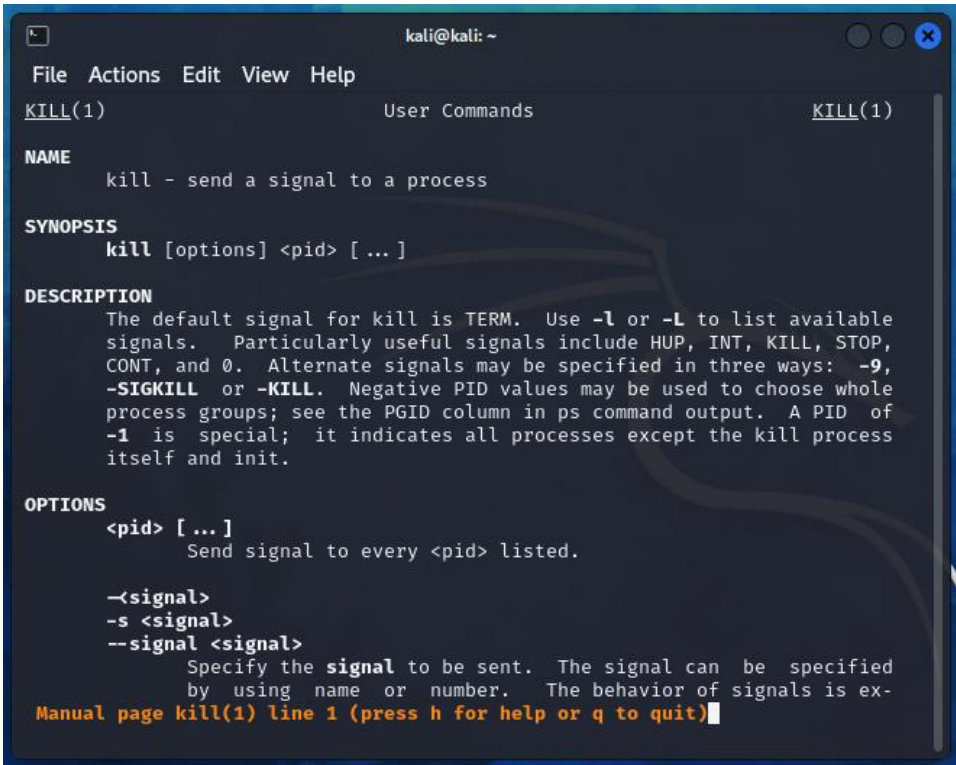
After 5 minute

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1073	kali	20	0	890288	127616	85128	S	0.7	6.3	0:27.86	xfwm4
1139	kali	20	0	297172	50356	23168	S	0.7	2.5	0:24.29	wrapper-2.0
1070	kali	20	0	155452	3716	3460	S	0.3	0.2	0:00.29	gpg-agent
1141	kali	20	0	274016	28744	21552	S	0.3	1.4	0:18.04	wrapper-2.0
31849	root	20	0	0	0	0	I	0.3	0.0	0:01.74	kworker/1:0-events
34005	kali	20	0	581728	66876	49844	S	0.3	3.3	0:06.85	qterminal
34048	kali	20	0	10476	5792	3616	R	0.3	0.3	0:03.09	top
1	root	20	0	23828	14684	10652	S	0.0	0.7	0:02.15	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.03	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u8:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:01.00	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:04.34	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.05	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.37	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1

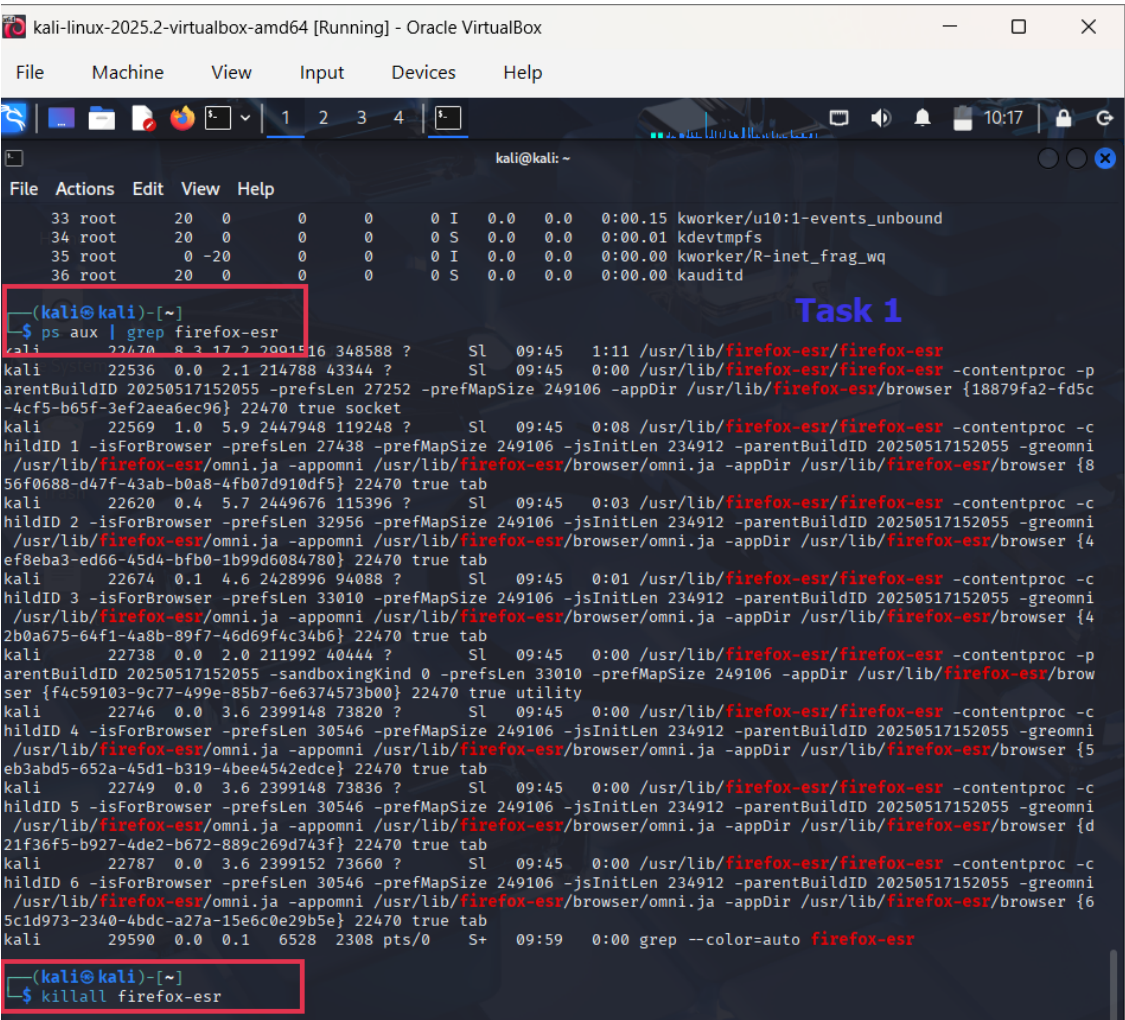
Observation after 5 minutes: Some processes spike, others drop when idle, and system background processes usually consume steady resources.

3. **Process Termination:** (Use with caution, recommended with `sudo` for system processes)

- The `kill` command allows you to terminate a process. Identify the process ID (PID) from the `ps` output. Use `kill <PID>` to terminate a process gracefully.
- For more forceful termination, use `kill -9 <PID>`.



Task 1: (Caution: This might affect ongoing tasks) Terminate a specific process using its name (e.g., `killall firefox`).



Task 2: Forcefully terminate a process that is unresponsive (use with extreme caution).

As you can see the process terminate politely.

```
(kali@kali)-[~]
$ killall firefox-esr

(kali@kali)-[~]
$ ps aux | grep firefox-esr
kali      30174  0.0  0.1  6528  2256 pts/0    S+   10:00   0:00 grep --color=auto firefox-esr
```

As you can see the process terminate politely.

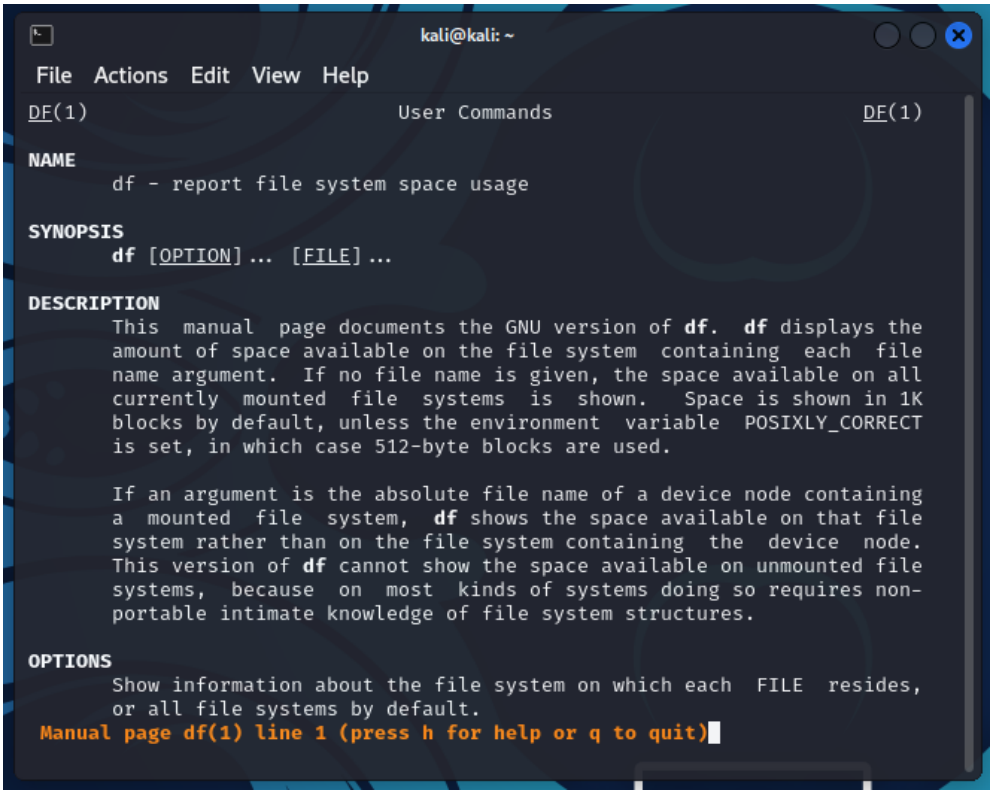
```
(kali@kali)-[~]
$ sudo killall -9 firefox-esr
firefox-esr: no process found

(kali@kali)-[~]
$
```

No action is needed because there is currently no Firefox process running.

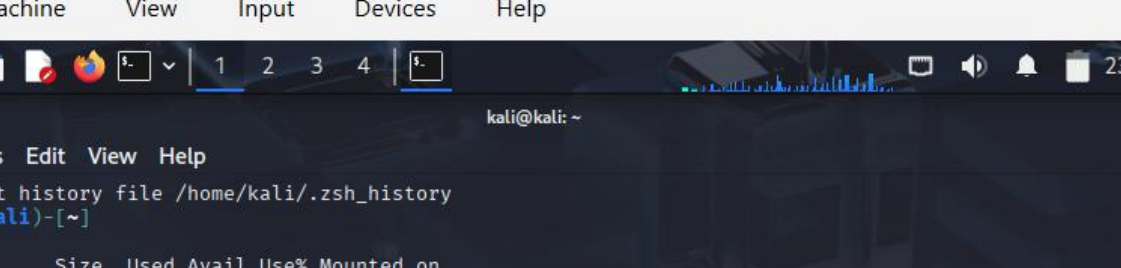
Part 2: Disk Space Management

4. **Disk Usage:** Use the `df` command to view information about mounted file systems. The `-h` option displays output in human-readable format (e.g., `df -h`). Analyze the available and used disk space for different partitions.



```
kali@kali: ~  
File Actions Edit View Help  
DF(1) User Commands DF(1)  
  
NAME  
df - report file system space usage  
  
SYNOPSIS  
df [OPTION]... [FILE]...  
  
DESCRIPTION  
This manual page documents the GNU version of df. df displays the  
amount of space available on the file system containing each file  
name argument. If no file name is given, the space available on all  
currently mounted file systems is shown. Space is shown in 1K  
blocks by default, unless the environment variable POSIXLY_CORRECT  
is set, in which case 512-byte blocks are used.  
  
If an argument is the absolute file name of a device node containing  
a mounted file system, df shows the space available on that file  
system rather than on the file system containing the device node.  
This version of df cannot show the space available on unmounted file  
systems, because on most kinds of systems doing so requires non-  
portable intimate knowledge of file system structures.  
  
OPTIONS  
Show information about the file system on which each FILE resides,  
or all file systems by default.  
Manual page df(1) line 1 (press h for help or q to quit)
```


Task 1: Identify the file system with the least available disk space.

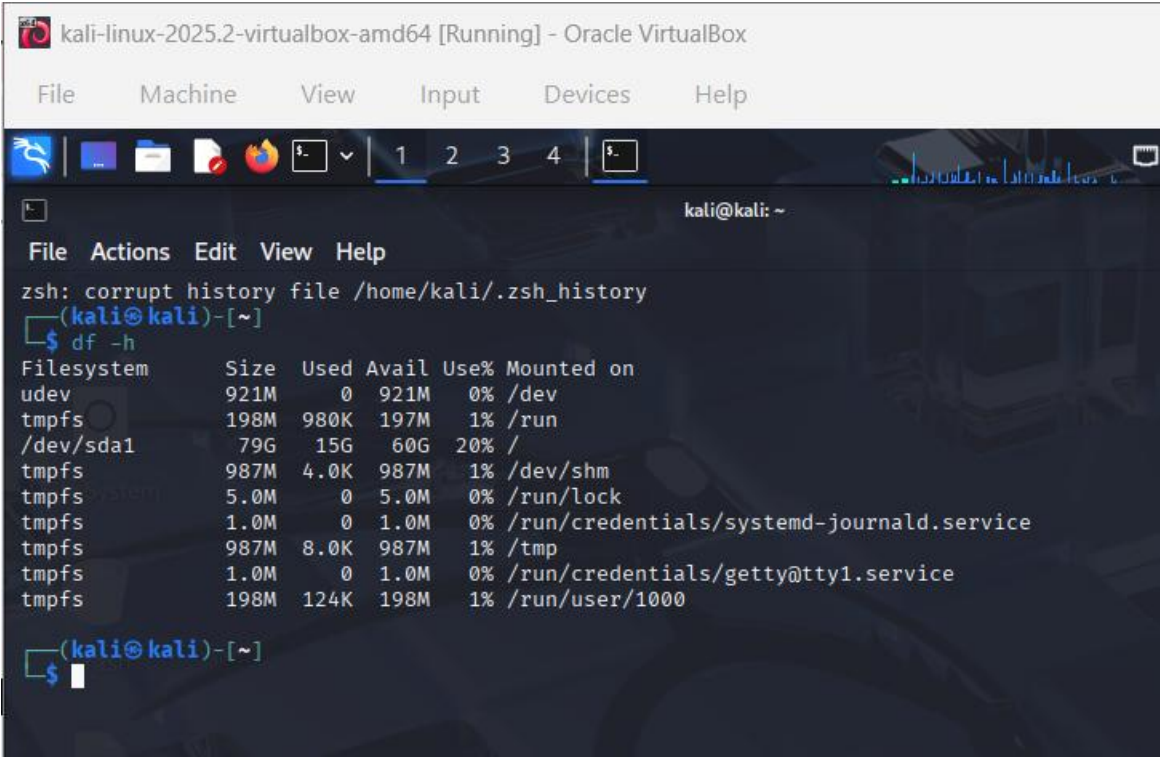


```
kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            921M   0    921M   0% /dev
tmpfs           198M  980K  197M   1% /run
/dev/sda1       79G   15G   60G  20% /
tmpfs           987M  4.0K  987M   1% /dev/shm
tmpfs           5.0M   0   5.0M   0% /run/lock
tmpfs           1.0M   0   1.0M   0% /run/credentials/systemd-journald.service
tmpfs           987M  8.0K  987M   1% /tmp
tmpfs           1.0M   0   1.0M   0% /run/credentials/getty@tty1.service
tmpfs          198M  124K  198M   1% /run/user/1000

(kali@kali)-[~]
$
```

Task 2: Analyze disk usage for different file systems and identify potential bottlenecks.



Observations:

1. Primary disk partition (/dev/sda1):

Only 20% used (15G of 79G), so no immediate disk bottleneck.
Plenty of available space (60G), so the root filesystem is not a concern.

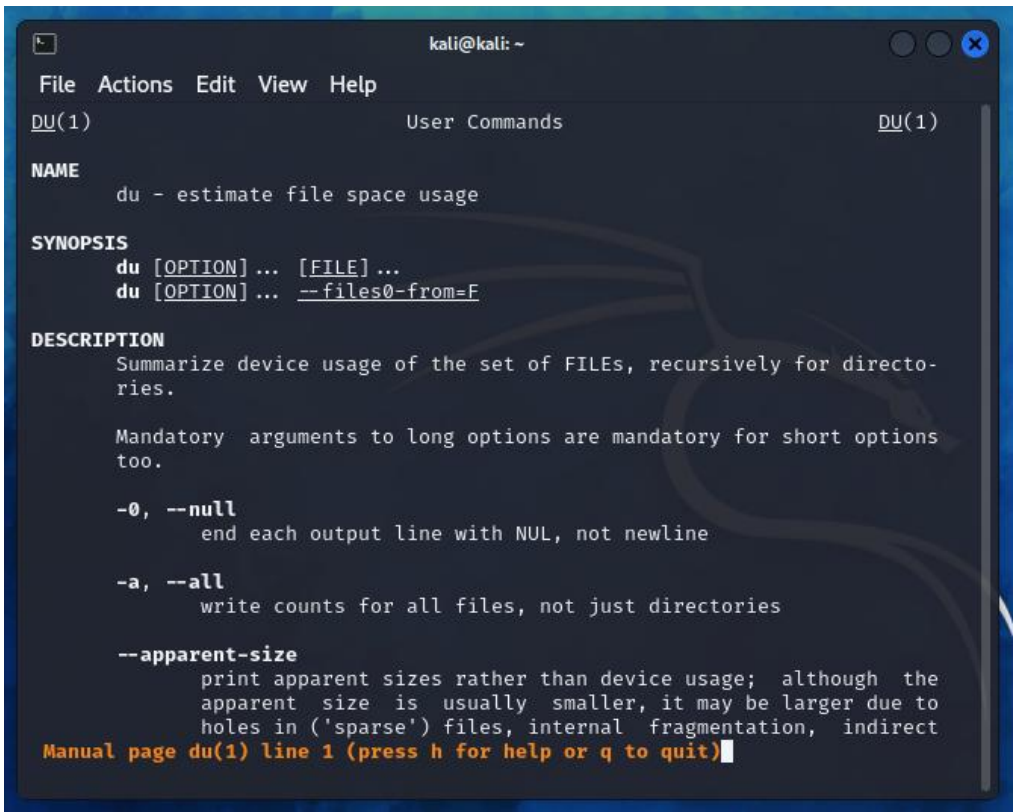
2. tmpfs and udev:

All tmpfs partitions and udev are low usage (<1–1%), as expected for temporary and in-memory filesystems.
These are unlikely to cause bottlenecks unless specific applications consume excessive temporary memory.

Potential Bottlenecks:

Currently, there are no disk usage bottlenecks. All partitions have ample free space. If /dev/sda1 usage approaches 80–90%, it could become a bottleneck for system operations. But no immediate issues. The system has plenty of free disk space, and the tmpfs partitions are not heavily used.

5. **Directory Usage:** Use the `du` command to estimate file and directory sizes. The `-h` option provides human-readable output and `-s` summarizes total disk usage for a directory (e.g., `du -sh /var/log`). Identify directories consuming significant disk space.



```
kali@kali: ~  
File Actions Edit View Help  
DU(1) User Commands DU(1)  
  
NAME  
    du - estimate file space usage  
  
SYNOPSIS  
    du [OPTION] ... [FILE] ...  
    du [OPTION] ... --files0-from=F  
  
DESCRIPTION  
    Summarize device usage of the set of FILES, recursively for directories.  
  
    Mandatory arguments to long options are mandatory for short options too.  
  
    -0, --null  
        end each output line with NUL, not newline  
  
    -a, --all  
        write counts for all files, not just directories  
  
    --apparent-size  
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect  
Manual page du(1) line 1 (press h for help or q to quit)
```


Task 1: Find the 3 largest files in your home directory.

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ du -ah ~ | sort -rh | head -n 3

108M    /home/kali
70M     /home/kali/.cache
66M     /home/kali/.cache/mozilla/firefox/9k6g45zo.default-esr

(kali@kali)-[~]
$
```

Task 2: Identify the directory within a specific path that consumes the most disk space.

```
(kali@kali)-[~]
$ du -ah ~ | sort -rh | head -n 3
108M    /home/kali
70M     /home/kali/.cache
66M     /home/kali/.cache/mozilla/firefox/9k6g45zo.default-esr

(kali@kali)-[~]
$ du -sh /home/kali/* | sort -rh | head -n 1
600K    /home/kali/Pictures

(kali@kali)-[~]
$
```

Task 3 (Optional): Consider learning about writing a script that recursively scans a directory and generates a report of the top 10 largest files.

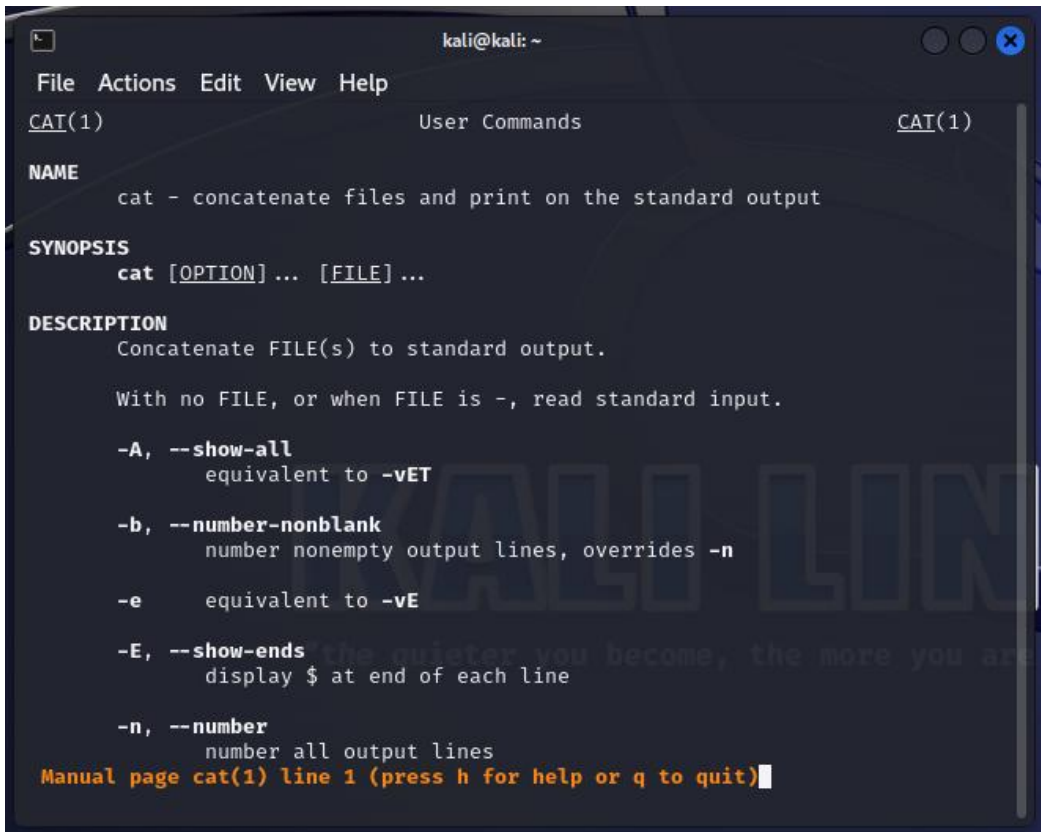
```
(kali@kali)-[~]
$ DIR=${1:-.}

(kali@kali)-[~]
$ echo "Scanning the top 10 Largest files $DIR"
Scanning the top 10 Largest files .

(kali@kali)-[~]
$ find "$DIR" -type f -exec du -h {} + 2>/dev/null | sort -rh | head -n 10
12M     ./mozilla/firefox/m1suuiou.default-esr/storage/permanent/chrome/idb/3870112724rsegmnoittet-es.sqlite
8.8M    ./cache/mozilla/firefox/m1suuiou.default-esr/safebrowsing/google4/goog-phish-proto.vlpset
8.6M    ./cache/mozilla/firefox/m1suuiou.default-esr/startupCache/scriptCache-current.bin
7.8M    ./cache/mozilla/firefox/m1suuiou.default-esr/cache2/entries/0D15A66668415866E5018198E50E947F6F261FE0
6.4M    ./cache/mozilla/firefox/m1suuiou.default-esr/cache2/entries/FCD17166F46B4F44FE682F70B937E7CCFE1348FE
5.0M    ./mozilla/firefox/m1suuiou.default-esr/places.sqlite
5.0M    ./mozilla/firefox/m1suuiou.default-esr/favicons.sqlite
4.9M    ./cache/mozilla/firefox/m1suuiou.default-esr/cache2/entries/B30AC03C95C3826EDC7EC782A62C4930672C6126
4.8M    ./cache/mozilla/firefox/m1suuiou.default-esr/cache2/entries/4R7428711DA64EB371CBAR0838D2R01315B6DAFD
4.7M    ./cache/mozilla/firefox/m1suuiou.default-esr/cache2/entries/B358FCE5EB28C1757A5C1C3FD346F5C004C03FDD
```

Part 3: Working with Data Files

6. **Viewing File Contents:** Use the `cat` command to display the contents of a text file. Be cautious using `cat` with large files as it can display everything at once.



```
kali@kali: ~  
File Actions Edit View Help  
CAT(1) User Commands CAT(1)  
  
NAME  
    cat - concatenate files and print on the standard output  
  
SYNOPSIS  
    cat [OPTION] ... [FILE] ...  
  
DESCRIPTION  
    Concatenate FILE(s) to standard output.  
  
    With no FILE, or when FILE is -, read standard input.  
  
    -A, --show-all  
        equivalent to -vET  
  
    -b, --number-nonblank  
        number nonempty output lines, overrides -n  
  
    -e      equivalent to -vE  
  
    -E, --show-ends  
        display $ at end of each line  
  
    -n, --number  
        number all output lines  
Manual page cat(1) line 1 (press h for help or q to quit)
```

Task 1: Display only the first 10 lines of a file.

```
(kali㉿kali)-[~/Music/Pop Rock/Sabrina Carpenter]
```

```
$ head -n 10 vampire.txt
```

```
Olivia Rodrigo - Vampire (Lyrics)
```

```
I hate to give the satisfaction askin' how you're doin' now  
How's the castle built off people you pretend to care about?  
Just what you wanted  
Look at you, cool guy, you got it  
I see the parties and the diamonds sometimes when I close my eyes  
Six months of torture you sold as some forbidden paradise  
I loved you truly  
You gotta laugh at the stupidity
```

10 lines

```
(kali㉿kali)-[~/Music/Pop Rock/Sabrina Carpenter]
```

```
$
```


Task 2: Combine the contents of two files into a single file.

```
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ cd Documents

(kali@kali)-[~/Documents]
$ cat message1.txt
cat: message1.txt: No such file or directory

(kali@kali)-[~/Documents]
$ nano message1.txt

(kali@kali)-[~/Documents]
$ cat message1.txt
Naruto Uzamaki

(kali@kali)-[~/Documents]
$ nano message2.txt

(kali@kali)-[~/Documents]
$ cat message2.txt
Sasuke Uchiha

(kali@kali)-[~/Documents]
$ cat message1.txt message2.txt > combined.txt

(kali@kali)-[~/Documents]
$ cat combined.txt
Naruto Uzamaki
Sasuke Uchiha

(kali@kali)-[~/Documents]
$
```

combined text

Task 3 (Optional) : Consider learning about writing a script that reads a file line by line and removes any empty lines.

```
(kali@kali)-[~/Documents]
$ cat message1.txt
Naruto Uzamaki

Konoha

(kali@kali)-[~/Documents]
$ cat message2.txt
Sasuke Uchiha

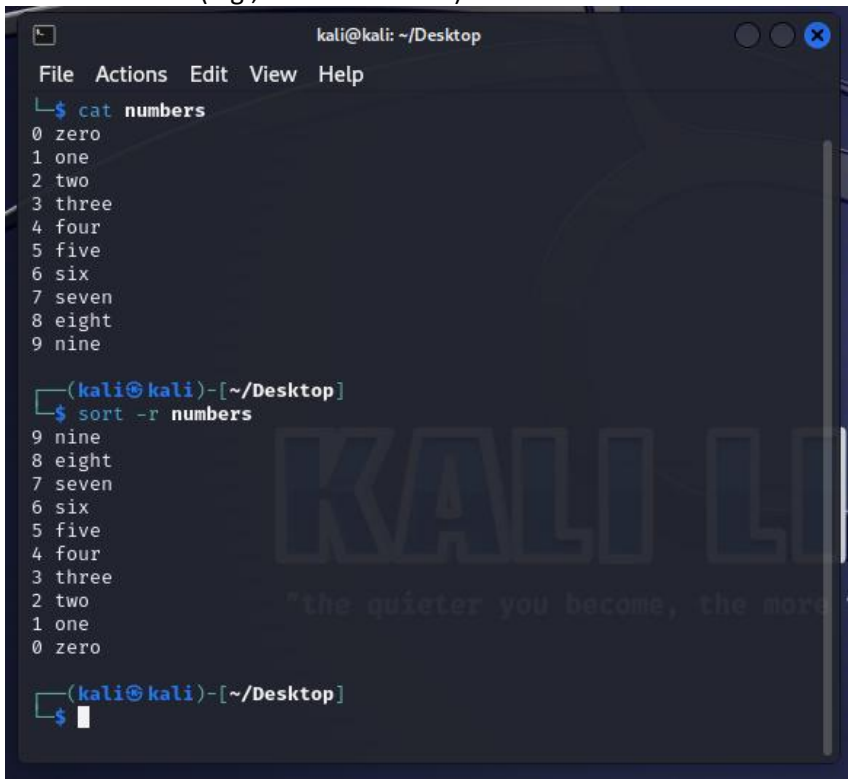
Hidden Leaf

(kali@kali)-[~/Documents]
$ grep -v '^$' message1.txt > message2.txt

(kali@kali)-[~/Documents]
$ cat message2.txt
Naruto Uzamaki
Konoha

(kali@kali)-[~/Documents]
$
```

7. **Sorting Data:** Use the `sort` command to arrange the lines of a text file in a specific order. The `-n` option sorts numerically, and `-r` sorts in reverse order (e.g., `sort -nr data.txt`).

A terminal window titled 'kali@kali: ~/Desktop' with a menu bar (File, Actions, Edit, View, Help). The first command is '\$ cat numbers', which outputs a list of numbers from 0 to 9 with their corresponding words: 0 zero, 1 one, 2 two, 3 three, 4 four, 5 five, 6 six, 7 seven, 8 eight, 9 nine. The second command is '\$ sort -r numbers', which outputs the same list in reverse order: 9 nine, 8 eight, 7 seven, 6 six, 5 five, 4 four, 3 three, 2 two, 1 one, 0 zero. The prompt '(kali@kali)-[~/Desktop]' is visible above each command.

```
kali@kali: ~/Desktop
File Actions Edit View Help
$ cat numbers
0 zero
1 one
2 two
3 three
4 four
5 five
6 six
7 seven
8 eight
9 nine

(kali@kali)-[~/Desktop]
$ sort -r numbers
9 nine
8 eight
7 seven
6 six
5 five
4 four
3 three
2 two
1 one
0 zero

(kali@kali)-[~/Desktop]
$
```

Task 1: Sort a file alphabetically based on the second column.

kali-linux-2025.2-virtualbox-amd64 [Running] - Oracle VirtualBox

FileMachineViewInputDevicesHelp

FileActionsEditViewHelp

zsh: corrupt history file /home/kali/.zsh_history

(kali@kali)-[~]

\$ cd Desktop

(kali@kali)-[~/Desktop]

\$ cat data1.txt

101 apple

102 orange

103 banana

104 grapes

105 mango

(kali@kali)-[~/Desktop]

\$ sort -k2,2 data1.txt

101 apple

103 banana

104 grapes

105 mango

102 orange

(kali@kali)-[~/Desktop]

\$

File System

Network

Browse Network

sort out by alphabetical order

Task 2: Sort a file numerically in descending order and display only the unique entries.

```
(kali@kali)-[~/Desktop]
$ cat data1.txt
101 apple
102 orange
103 banana
104 grapes
105 mango

(kali@kali)-[~/Desktop]
$ sort -nr data1.txt | uniq
105 mango
104 grapes
103 banana
102 orange
101 apple

(kali@kali)-[~/Desktop]
$
```

sort out descending order

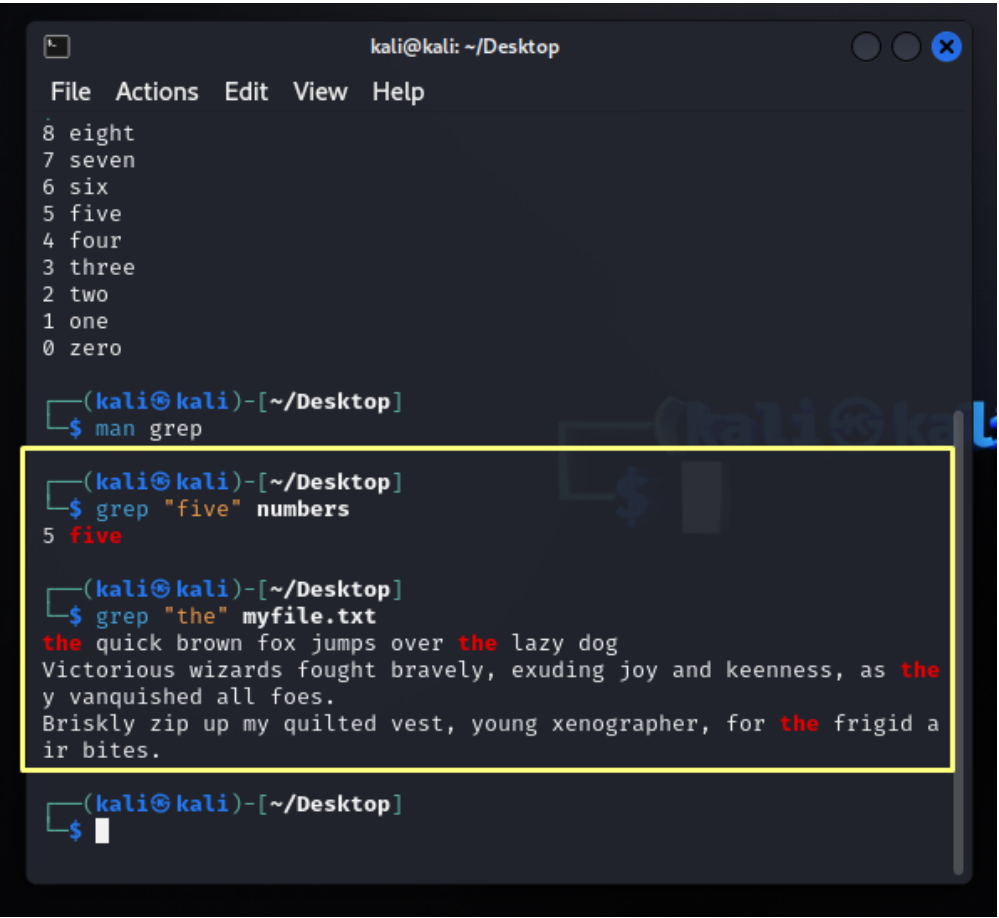
Task 3 (Optional) : Consider learning about writing a script that sorts a log file by date and extracts specific information from each line.

```
(kali@kali)-[~/Desktop]
$ awk '{print $1, $3}' abc.txt
2025-09-21 INFO
2025-09-21 ERROR
2025-09-21 INFO

(kali@kali)-[~/Desktop]
$ cat abc.txt
2025-09-21 15:32:10 INFO User paul logged in
2025-09-21 15:35:42 ERROR Database connection failed
2025-09-21 15:36:00 INFO User aamada logged out

(kali@kali)-[~/Desktop]
$ awk '{print $1, $3 "secret" "$5" }' abc.txt
2025-09-21 INFOsecret$5
2025-09-21 ERRORsecret$5
2025-09-21 INFOsecret$5
```

8. **Searching Text:** Use the `grep` command to search for specific patterns within a text file. The basic syntax is `grep <pattern> <filename>`. Explore options like `-i` for case-insensitive search and `-c` to count occurrences of the pattern.



```
kali@kali: ~/Desktop
File Actions Edit View Help
8 eight
7 seven
6 six
5 five
4 four
3 three
2 two
1 one
0 zero

(kali@kali)-[~/Desktop]
$ man grep

(kali@kali)-[~/Desktop]
$ grep "five" numbers
5 five

(kali@kali)-[~/Desktop]
$ grep "the" myfile.txt
the quick brown fox jumps over the lazy dog
Victorious wizards fought bravely, exuding joy and keenness, as the
y vanquished all foes.
Briskly zip up my quilted vest, young xenographer, for the frigid a
ir bites.

(kali@kali)-[~/Desktop]
$
```

Task 1: Search for a specific email address within a file and display the entire line containing the address.

```
(kali@kali)-[~/Desktop]
$ cd ~/Documents
(kali@kali)-[~/Documents]
$ nano email.txt
(kali@kali)-[~/Documents]
$ cat email.txt
alice.smith@email.com

john.doe123@mail.com

linda.green2025@gmail.com

mike_brown@outlook.com

sarah.jones99@yahoo.com

tom.wilson-office@company.com

emma.thompson+test@gmail.com

chris.adams_dev@mail.com

olivia.white2025@outlook.com

ryan.miller1234@yahoo.com

(kali@kali)-[~/Documents]
$ grep "linda.green2025@gmail.com" emai.txt
grep: emai.txt: No such file or directory

(kali@kali)-[~/Documents]
$ grep "linda.green2025@gmail.com" email.txt
linda.green2025@gmail.com

(kali@kali)-[~/Documents]
$
```

Task 2: Count the number of lines in a file that contain the word “error”.

```
linda.green2025@gmail.com

(kali@kali)-[~/Documents]
$ grep -c "error" email.txt
0

(kali@kali)-[~/Documents]
$
```

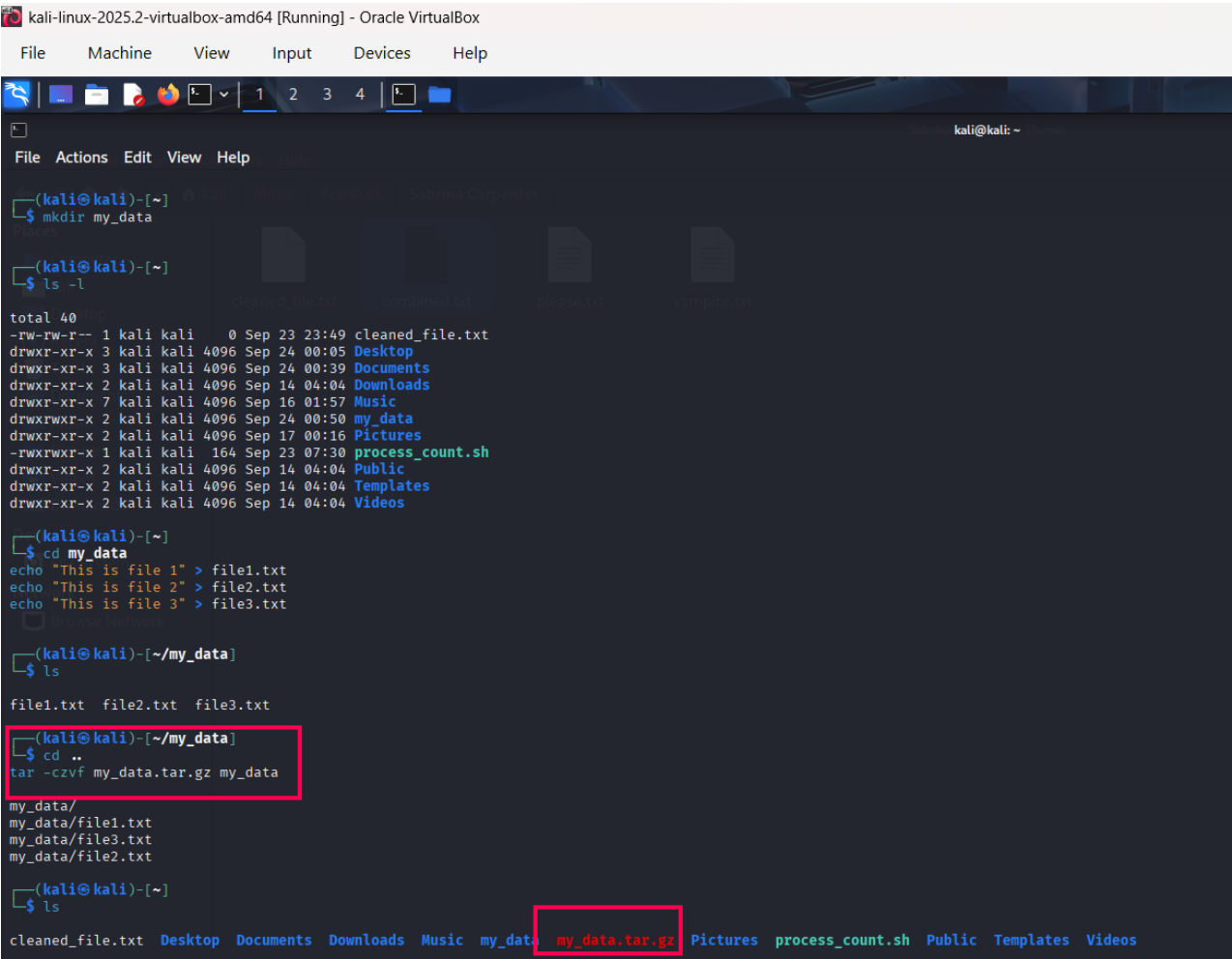
There are zero error in my “email.txt”

Task 3 (Optional): Consider learning about writing a script that searches for a specific pattern in multiple files within a directory.

Part 4: Data Compression and Archiving

9. **Compressing Files:** Use the `gzip` command to compress individual files, reducing their size. The compressed file will have a `.gz` extension (e.g., `gzip data.txt`).

Task 1: Compress a directory and all its contents into a single archive file.



Task 2: Decompress a specific file within a compressed archive without extracting the entire archive.

```
(kaliⓈkali)-[~]  
$ tar -xzvf my_data.tar.gz my_data/file2.txt  
my_data/file2.txt  
  
(kaliⓈkali)-[~]  
$ ls my_data  
file1.txt  file2.txt  file3.txt  
  
(kaliⓈkali)-[~]  
$
```

Decompress a specific file

Task 3 (Optional): Consider learning about writing a script that automatically compresses all files with a specific extension within a directory.

```
(kali@kali)-[~]
$ find Music/ -type f -name -print0 | tar -cvzf Music.tar.gz --null -T -

(kali@kali)-[~]
$ find Music/ -type f -name "Time and Tides.txt" -print0 | tar -cvzf Music.tar.gz --null -T -
Music/Ballad/Basia/Time and Tides.txt

(kali@kali)-[~]
$
```

kali@kali: ~/Desktop

FileActionsEditViewHelp

GZIP(1)General Commands ManualGZIP(1)

NAME

gzip, gunzip, zcat - compress or expand files

SYNOPSIS

gzip [-acdfhklLnNrtvV19] [-S suffix] [name ...]
gunzip [-acfhklLnNrtvV] [-S suffix] [name ...]
zcat [-fhLV] [name ...]

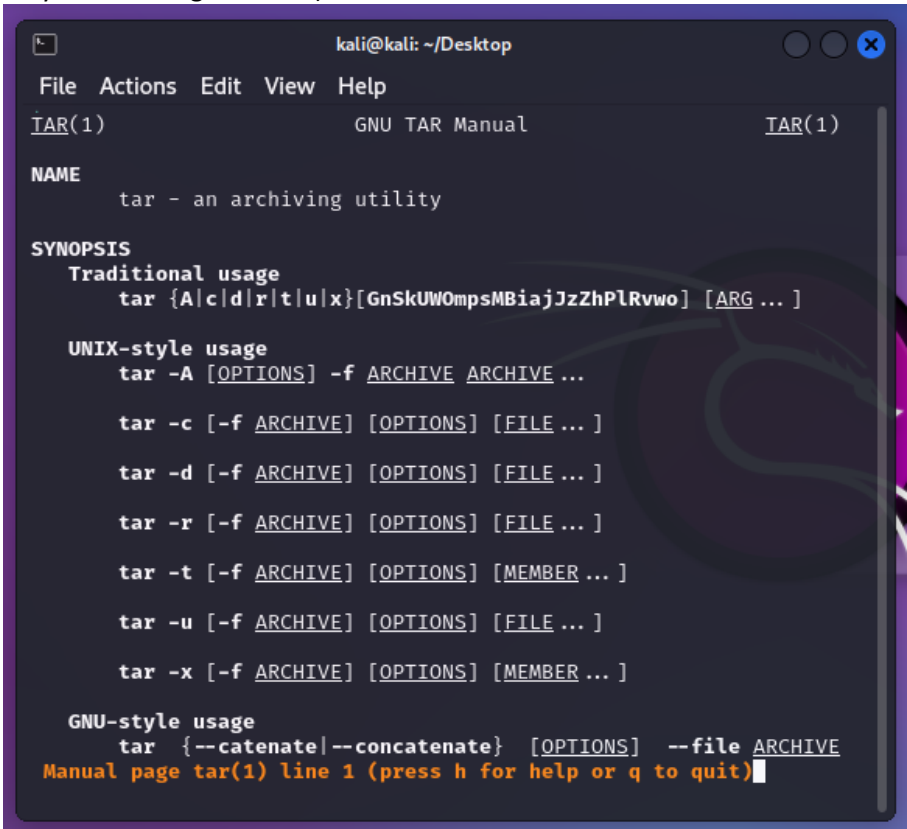
DESCRIPTION

The **gzip** command reduces the size of the named files using Lempel-Ziv coding (LZ77). Whenever possible, each file is replaced by one with the extension **.gz**, while keeping the same ownership modes, access and modification times. (The default extension is **z** for MSDOS, OS/2 FAT, Windows NT FAT and Atari.) If no files are specified, or if a file name is "-", the standard input is compressed to the standard output. The **gzip** command will only attempt to compress regular files. In particular, it will ignore symbolic links.

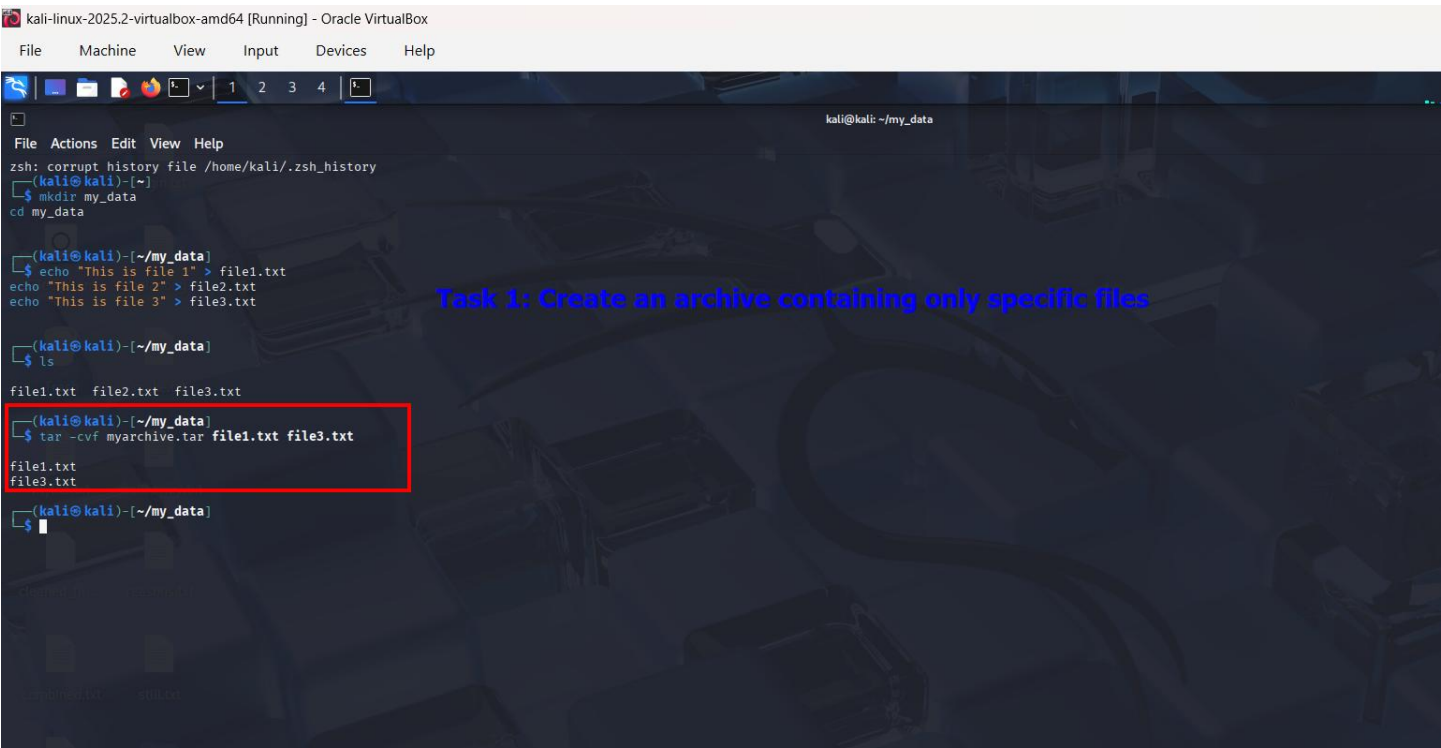
If the compressed file name is too long for its file system, **gzip** truncates it. The **gzip** command attempts to truncate only the parts of the file name longer than 3 characters. (A part is delimited by dots.) If the name

Manual page gzip(1) line 1 (press h for help or q to quit)

10. **Archiving Files:** Use the `tar` command to create archive files, typically with a `.tar` extension. The `-cvf` options create a new archive file with the specified name (e.g., `tar -cvf myarchive.tar data.txt`). Explore options for compressing archives with `gzip` (e.g., `tar -cvzf myarchive.tar.gz data.txt`).



Task 1: Create an archive file containing only specific files from a directory.



Task 2: Extract specific files from an archive based on their names or patterns.

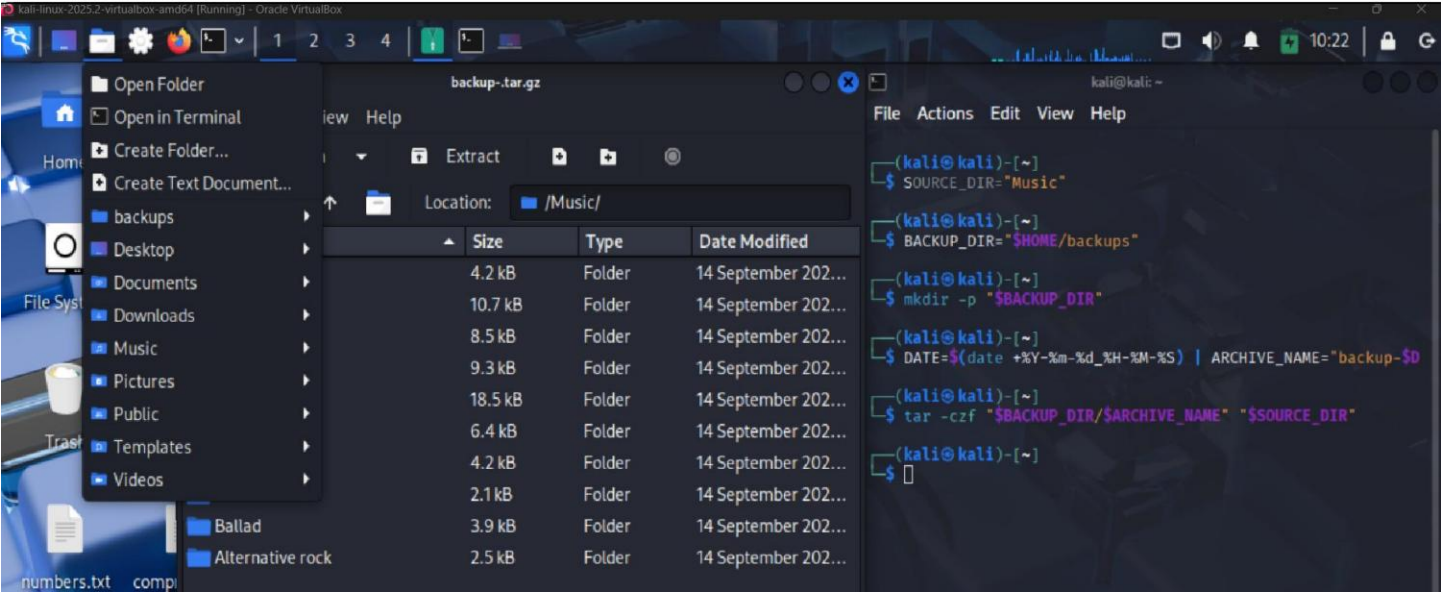
```
(kali@kali)-[~/my_data]
$ tar -xvf myarchive.tar file3.txt
file3.txt

(kali@kali)-[~/my_data]
$ ls
file1.txt file2.txt file3.txt myarchive.tar

(kali@kali)-[~/my_data]
$
```

Extract specific files from an archive
Only file3.txt is extracted to the current directory.

Task 3(Optional): Consider learning about writing a script that backs up a directory structure to a compressed archive file daily.



Further Exploration:

- Research additional options and functionalities of the covered commands.
- Explore commands for managing user accounts, file permissions, and network connections.

"Code, our weapon. Knowledge, our shield.
Together, we defend the digital frontier!"
- Kappattid